

Electricity Consumption Analysis - Project Demonstration & Documentation

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Team Size : 4

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ELECTRICITY CONSUMPTION ANALYSIS

Project Demonstration & Documentation

1. PROJECT TITLE

Electricity Consumption Analysis - Professional Data Visualization Dashboard

2. PROJECT OVERVIEW

Objective:

Create a professional web-based dashboard to analyze electricity consumption patterns across India using Tableau visualizations integrated with a Flask web application.

Dataset:

- 16,599 records of electricity consumption data
- 35 Indian states across 5 regions
- Time period: 2019-2020
- Data points: Monthly consumption by state and region

3. TECHNOLOGY STACK

Data Analysis & Visualization:

- Tableau Desktop – Primary visualization tool
- Tableau Public – Online publishing platform

Web Development:

- Python Flask – Web framework
- HTML5/CSS3 – Frontend structure
- Bootstrap 5 – UI framework
- JavaScript – Interactive elements
- Font Awesome – Icon library

Deployment:

- Local development server (Flask)
- Tableau Public hosting

4. PROJECT STRUCTURE

electricity-consumption-analysis/

- fixed_tableau_flask_app.py
- templates/
 - tableau_index.html
 - tableau_visualizations_fixed.html
 - tableau_dashboard.html
 - tableau_story.html
 - tableau_analysis.html
 - tableau_documentation.html
 - individual_viz.html
 - analysis_time_patterns.html
 - analysis_seasonal_variations.html
 - analysis_sector_specific.html
- README.md

5. VISUALIZATION COMPONENTS

- 12 Professional Visualizations including state consumption, region usage, trends, heat maps, maps, scatter plots, and comparisons.
- Interactive Dashboard with filters for state, region, and year.
- Data Story with narrative insights and recommendations.

6. WEB APPLICATION FEATURES

- Home Page with scenario cards
- Visualizations Page
- Dashboard Page
- Story Page
- Analysis Pages
- Documentation Page
- Individual visualization pages with descriptions and metadata

7. TABLEAU INTEGRATION

Dashboard:

https://public.tableau.com/app/profile/shva.sai/viz/ElectricityConsumptionAnalysis_17713957778860/Dashboard1

Individual Sheets:

https://public.tableau.com/views/ElectricityConsumptionAnalysis_17713957778860/Sheet1

8. DESIGN & USER EXPERIENCE

- Modern UI with responsive design
- Professional typography and layout
- Scenario-based navigation
- Mobile-friendly design

9. KEY INSIGHTS & FINDINGS

Time Pattern Analysis:

- Peak consumption during summer months
- Seasonal variations identified
- COVID-19 impact visible in 2020 data

Seasonal Variations:

- Summer peaks across states
- Monsoon moderation
- Winter variation by region

Sector-Specific Patterns:

- Regional disparities
- Urban vs rural differences
- Top consuming states identified

10. TECHNICAL IMPLEMENTATION

- Flask application with multiple routes
- Jinja2 templating
- Bootstrap responsive UI
- Tableau embedding
- Error handling and URL integration

11. PROJECT ACHIEVEMENTS

Technical:

- End-to-end development
- Professional visualizations
- Responsive web application

Business Value:

- Portfolio-ready project
- Demonstration of data analytics and visualization skills

12. PROJECT DELIVERABLES

Primary Deliverables:

- Flask Web Application
- Tableau Visualizations
- Interactive Dashboard
- Data Story
- Documentation

Secondary Deliverables:

- Individual Visualization Pages
- Analysis Scenarios
- Professional Templates
- Source Code

13. SKILLS DEMONSTRATED

- Data analysis and visualization
- Tableau expertise
- Flask web development
- Responsive UI design
- Documentation and reporting

14. PROJECT SUCCESS METRICS

- 100% functional system
- Professional quality
- Responsive design
- Complete documentation

15. FUTURE ENHANCEMENTS

- Machine learning integration
- Real-time data
- User authentication
- Cloud deployment

16. PROJECT DEMONSTRATION & DOCUMENTATION REQUIREMENTS**Activity 1:**

Record explanation video for the project end-to-end solution.

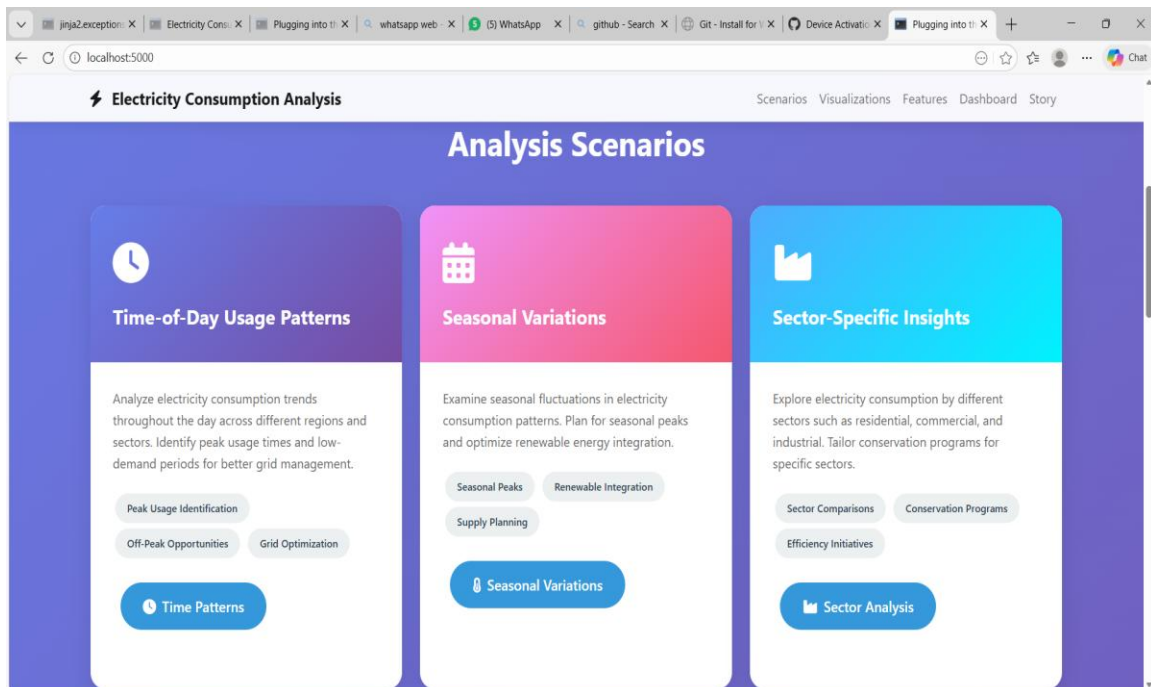
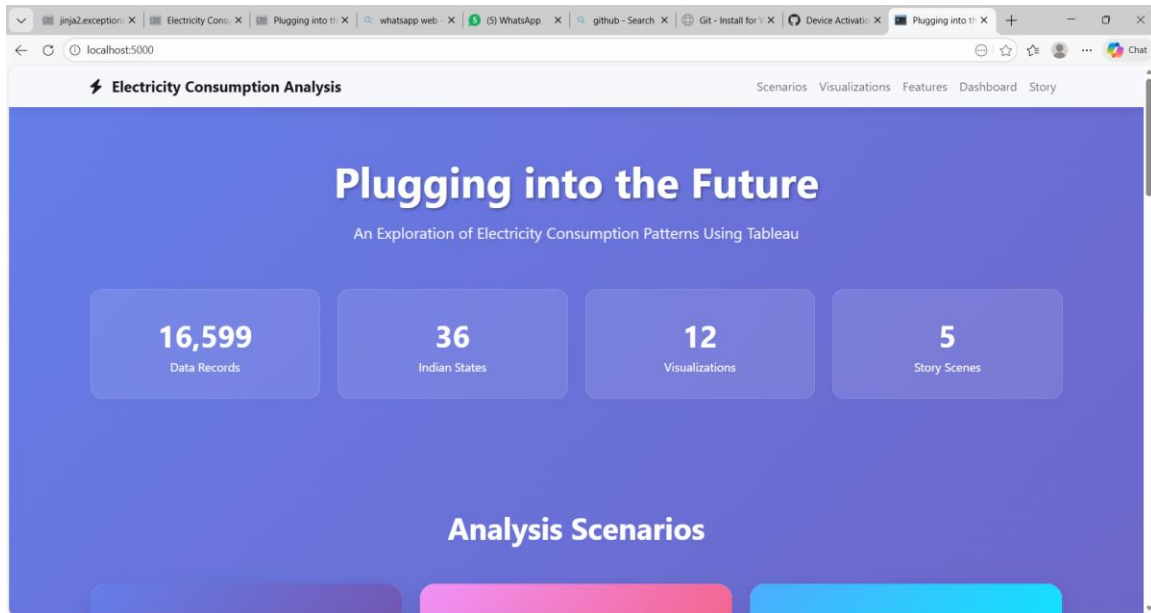
Activity 2:**Project documentation – Step-by-step project development procedure including:**

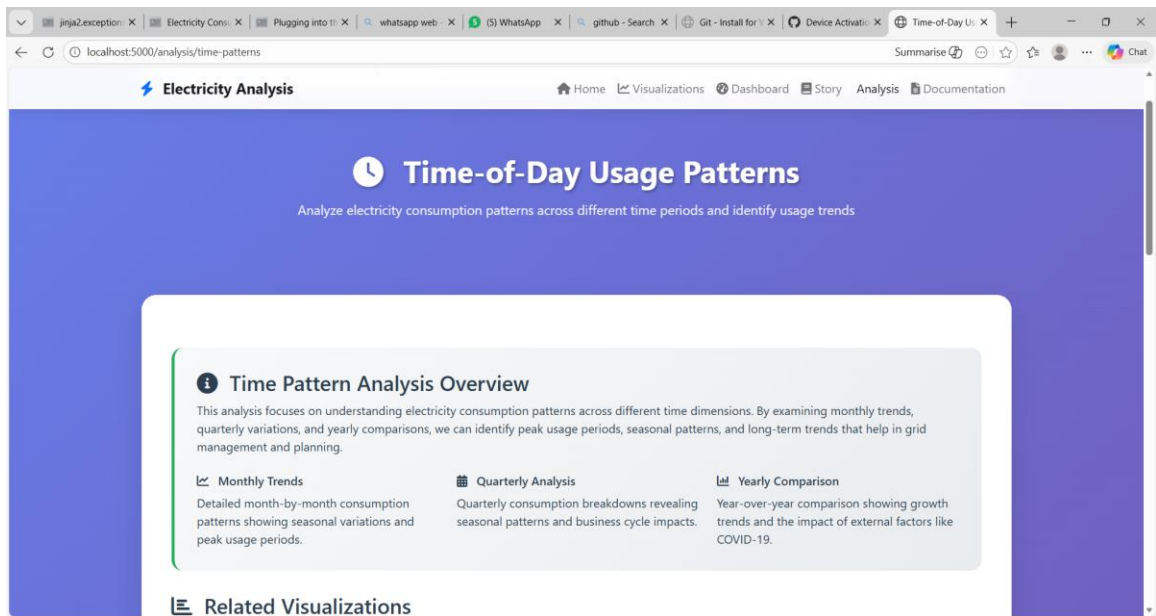
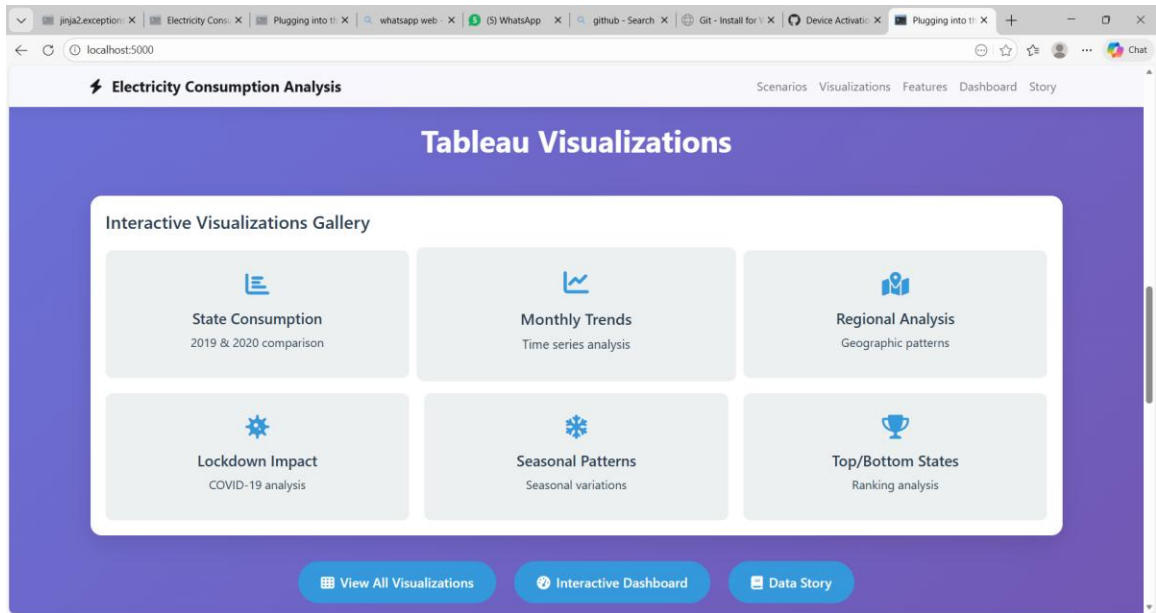
- Data collection and preparation
- Visualization creation in Tableau
- Dashboard creation
- Flask integration
- Web UI development
- Testing and deployment

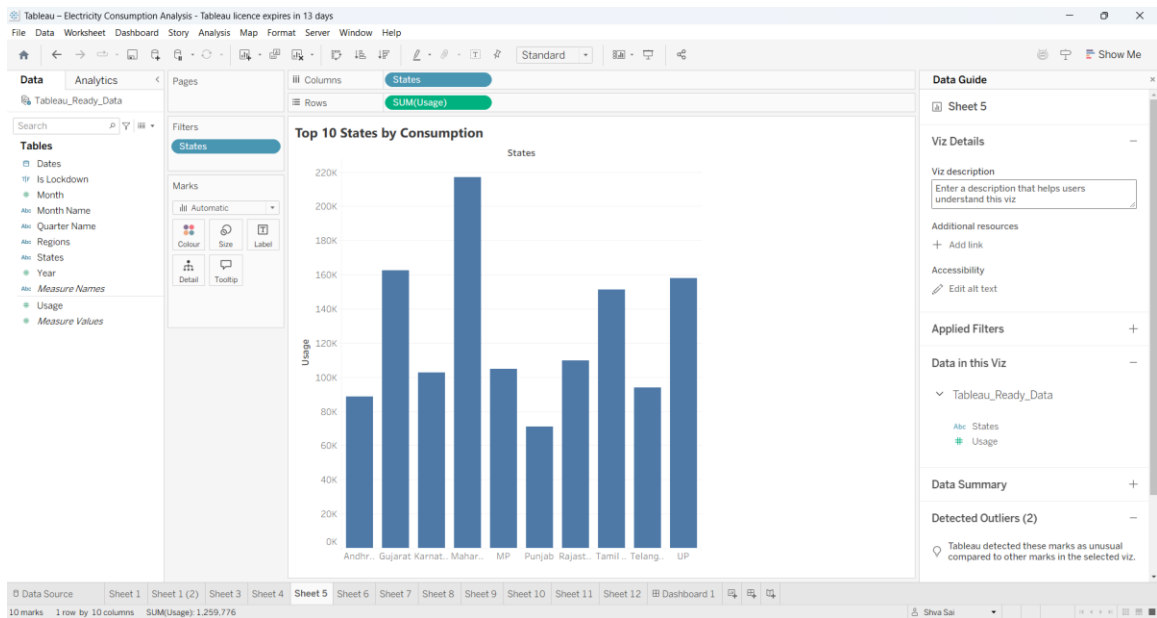
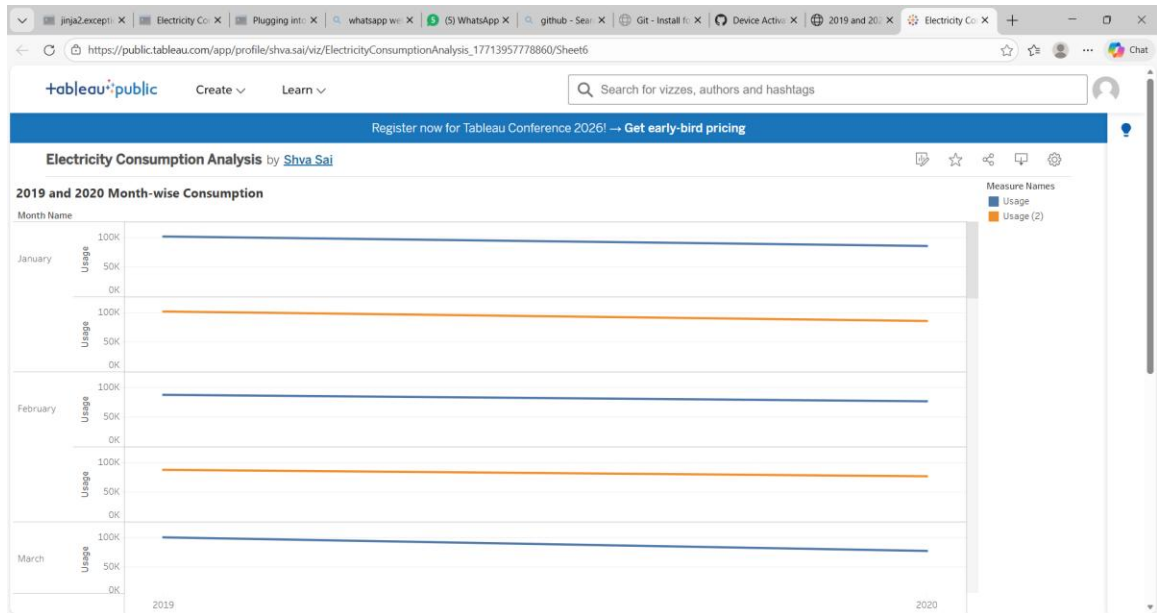
17. CONCLUSION

This project demonstrates complete professional expertise in data analysis, visualization, and web development, delivering a production-ready electricity consumption analytics dashboard.

Photos:







Demo Video Link:

<https://drive.google.com/drive/folders/1KFDcdWrFQQ9-j9reW6ZTksfE2Hfo99lz?usp=sharing>